NASA Space Apps Challenge, Cairo 2026

Thermora



Who lit it? Every fire seen from orbit, read as natural or human.

Team Zorgons

The problem

Two satellites, two different fire records

MODIS, since 2000

The long record. Pixels about 1 km wide, so small fires, such as crop burning, often go unseen.

VIIRS, since 2012

Pixels 375 m wide. It sees many of the small fires MODIS misses, but only for the last decade or so.

Read as one story, the long record undercounts fire, and neither says whether a fire was lit by people or by nature.

Our question

How much fire have we been missing, and how much of it did people light?

Our solution

Thermora puts both answers on one globe

1

Harmonize

Fuse MODIS and VIIRS detections into one fire record per place and day.

2

Measure the gap

Show, per country and year, how many more fires the harmonized record finds than MODIS alone.

3

Human or natural

Colour every area by the science team’s human-association score, with its 95% range and confidence.

A likelihood for an area, never a verdict on a single fire.

How it works

From satellite pixels to a living globe

Data engineering

Harmonize

MODIS and VIIRS matched into fire records on a common ~0.01° pixel, per date and pass.

Data science

Score

Each fire gets a 0–1 human-association score, a 95% interval and a confidence level.

Backend

Serve

ASP.NET and DuckDB join both tables by fire ID and serve countries, grids and calendars.

Frontend

Explore

A 3D globe with country, place and day views, a filter and a guided tour.

Both tables are Parquet files on Hugging Face; the API reads them directly, so there is no database to run.

The data

Two tables, joined by fire ID

Table	Made by	What we use
harmonized_data	Data engineer	Country, location, date, MODIS detections, all detections, fire type
fire_level_assessment	Data scientist	Human score, 95% lower and upper bound, confidence level

Angola · China · Egypt · Greece · United Kingdom · United States · Yemen

Seven countries, 2000 to 2025. Not every fire is scored, so some areas honestly have no score.

Measure 1

How much does the old record miss?

$$\text{Undercount} = (\text{harmonized} - \text{legacy}) \div \text{legacy}$$

Legacy

What MODIS alone recorded: the record most long-term studies use.

Harmonized

MODIS and VIIRS together: the fuller picture of the same places and days.

Computed for every country and year, and for any searched place within a radius.

Measure 2

Human or natural, with the uncertainty shown



Likely natural

Likely human-driven

Score

The average 0–1 score of the area's scored fires.

95% range

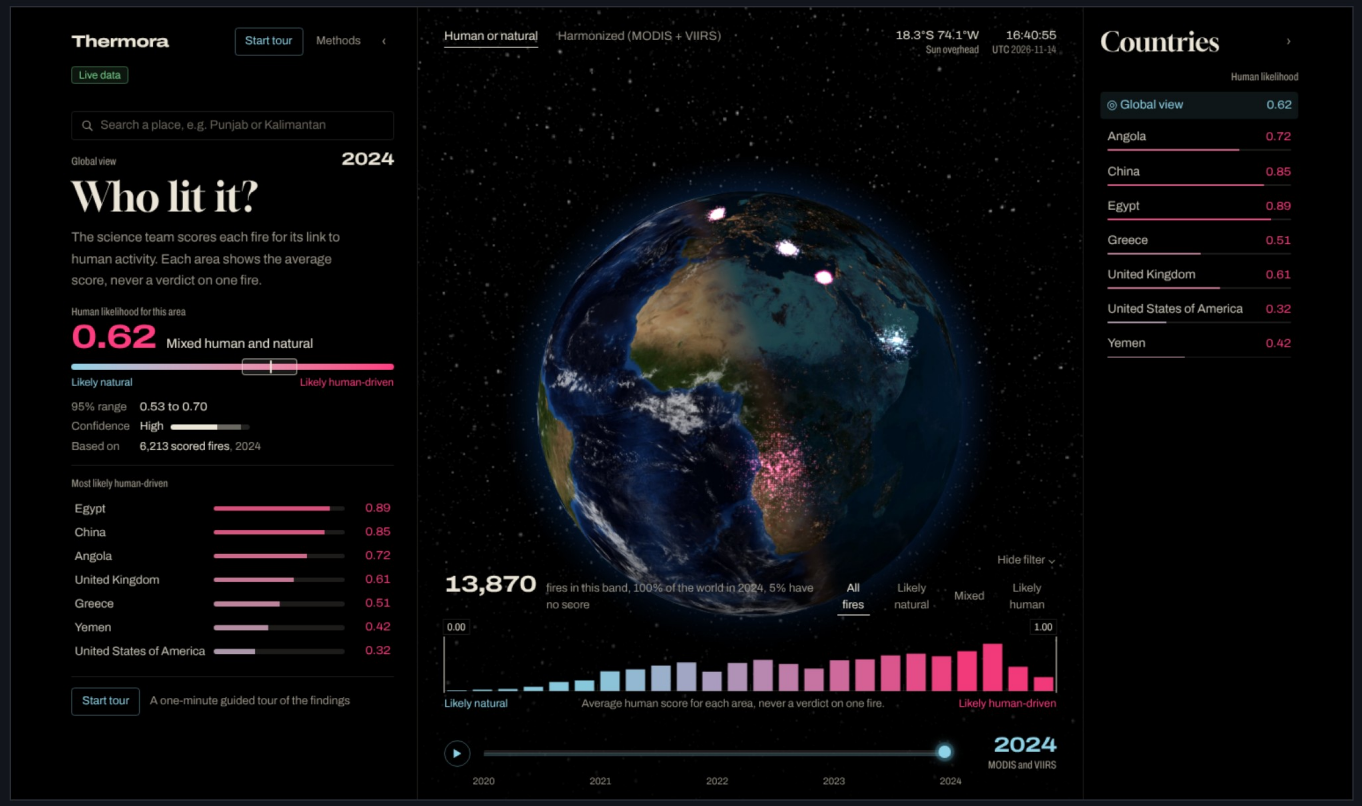
The bracket: the average of those fires' intervals.

Confidence

Low, medium or high, plus how many fires were scored.

The app

A globe that answers “who lit it?”



Interface shown on test data.

Two views

Human or natural, and harmonized MODIS + VIIRS.

A filter you can drag

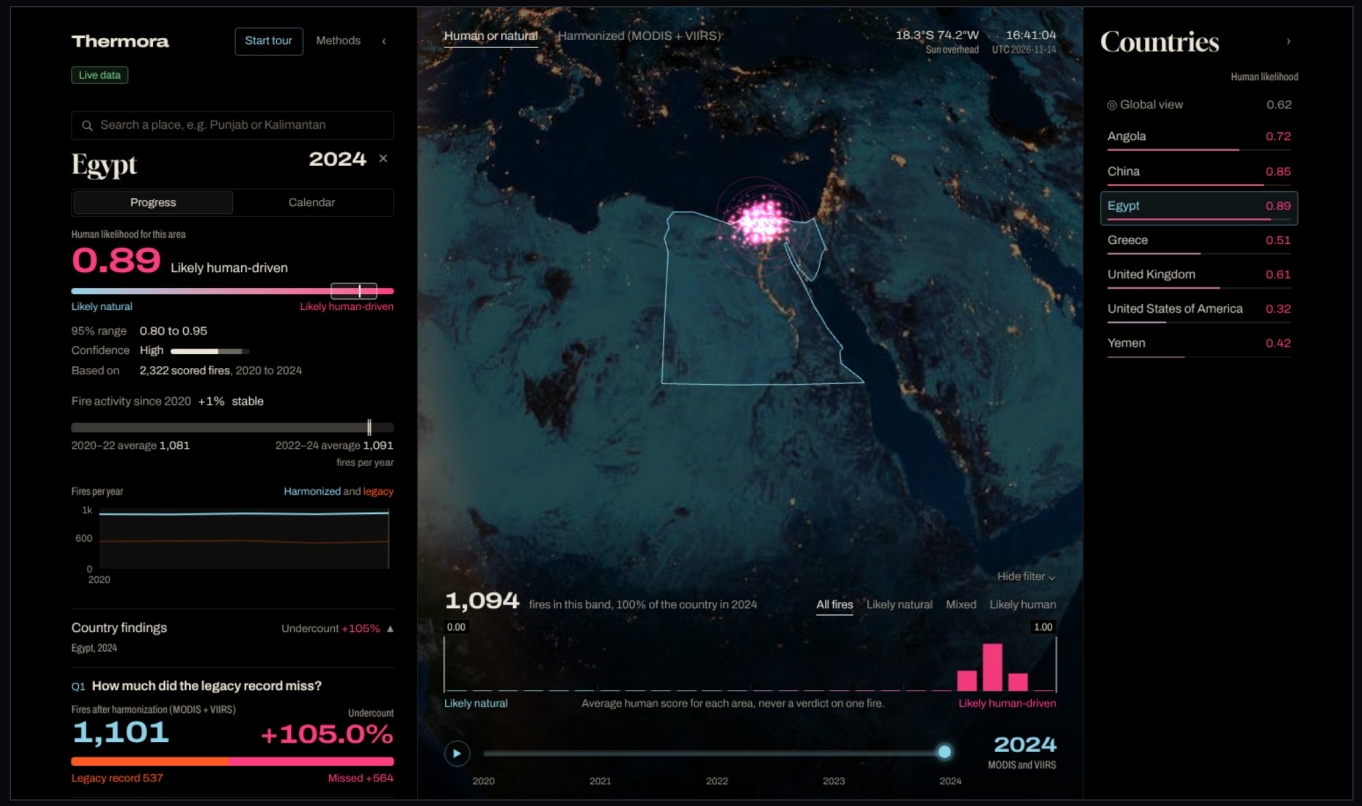
Show only likely-natural, mixed or likely-human areas.

Countries, ranked

One click flies the globe to a country.

The app

From a country down to a single day



Interface shown on test data.

Any place

Search a town or click the globe; fires are counted within a radius.

Since the start

Is fire activity getting worse or better there?

Calendar

Every day of the year, and how one day compares with others.

Guided tour

One minute, chosen from the data.

Principles

Honest by design

Real data only

Every number comes from the backend. No sample or placeholder figures, ever.

Uncertainty shown

Each human score carries its 95% range, confidence and how many fires it rests on.

Unknown stays unknown

Areas without scored fires are grey and say “no score”, never zero.

Says when it can’t

If the server is waking up or unreachable, the app says so instead of guessing.

Technology

What we built it with

Layer	What we used
Frontend	React 19, TypeScript, Vite, three.js with custom shaders, Tailwind, zustand
Globe	Live day and night lighting from the sun’s position, fire embers in one draw call
Backend	ASP.NET Core and DuckDB, parameterized SQL, 30-minute cache
Data	Parquet on Hugging Face, joined at query time by fire ID
Deploy	Vercel for the app, a Docker container for the API

What we found

Findings, from the live data

+[__]%	Worldwide in [year], the harmonized record finds [__] fires against [__] in MODIS alone.
[0.__]	[Country] scores most likely human-driven (95% range [__] to [__], [level] confidence).
[0.__]	[Country] scores most likely natural, with [__] fires scored.

Next steps

Where Thermora goes next

Sharper counts

Count fire records, not detections, and set industrial and volcanic sources apart from vegetation fires.

Truer ranges

A proper 95% interval for each area's average, not the average of per-fire intervals.

More of Earth

More countries and a faster first load, so anyone can ask "who lit it?" anywhere.

Team Zorgons

Thank you

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Try it: https://zorgons-thermora.vercel.app/

Code: github.com/T0-kyo/zorgons-thermora. Fire data: NASA FIRMS (MODIS and VIIRS).